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Envicool

Beyond Megawatts: Engineering Liquid Cooling for AI at Scale



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OCP GLOBAL SUMMIT

☑ Founded in **2005**, focus on thermal management;

☑ Global cooling solution provider for AI infrastructure, energy storage and industrial applications;

☑ Supporting customers from rapid prototyping through full-scale mass production;

☑ Visibility into several stages of the cooling system;

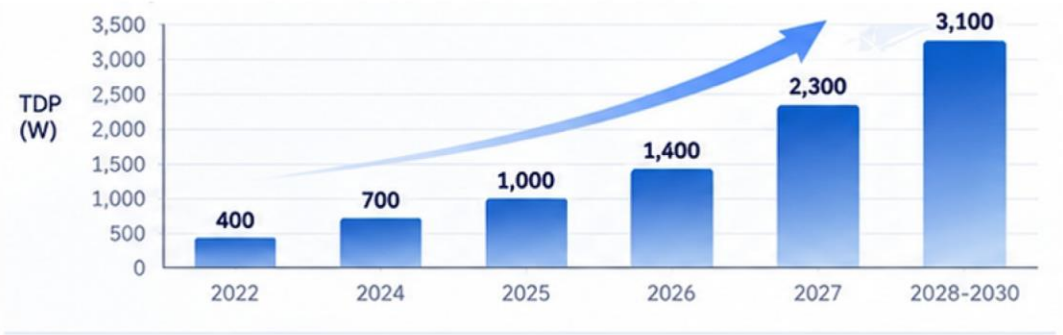
21
years experiences
in Thermal Mgmt.

9000+
Global Employee
30% R&D

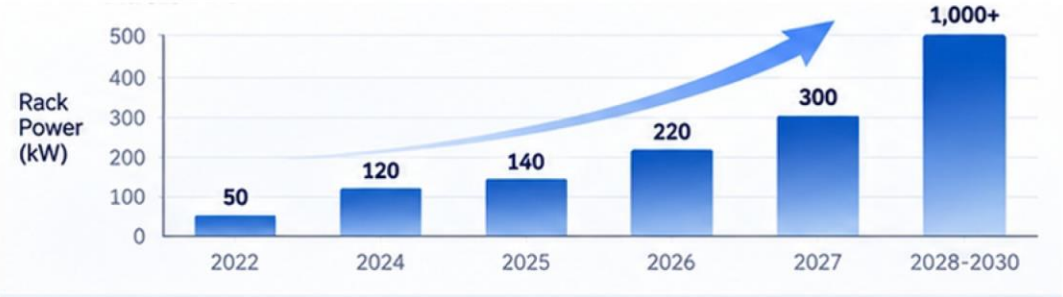


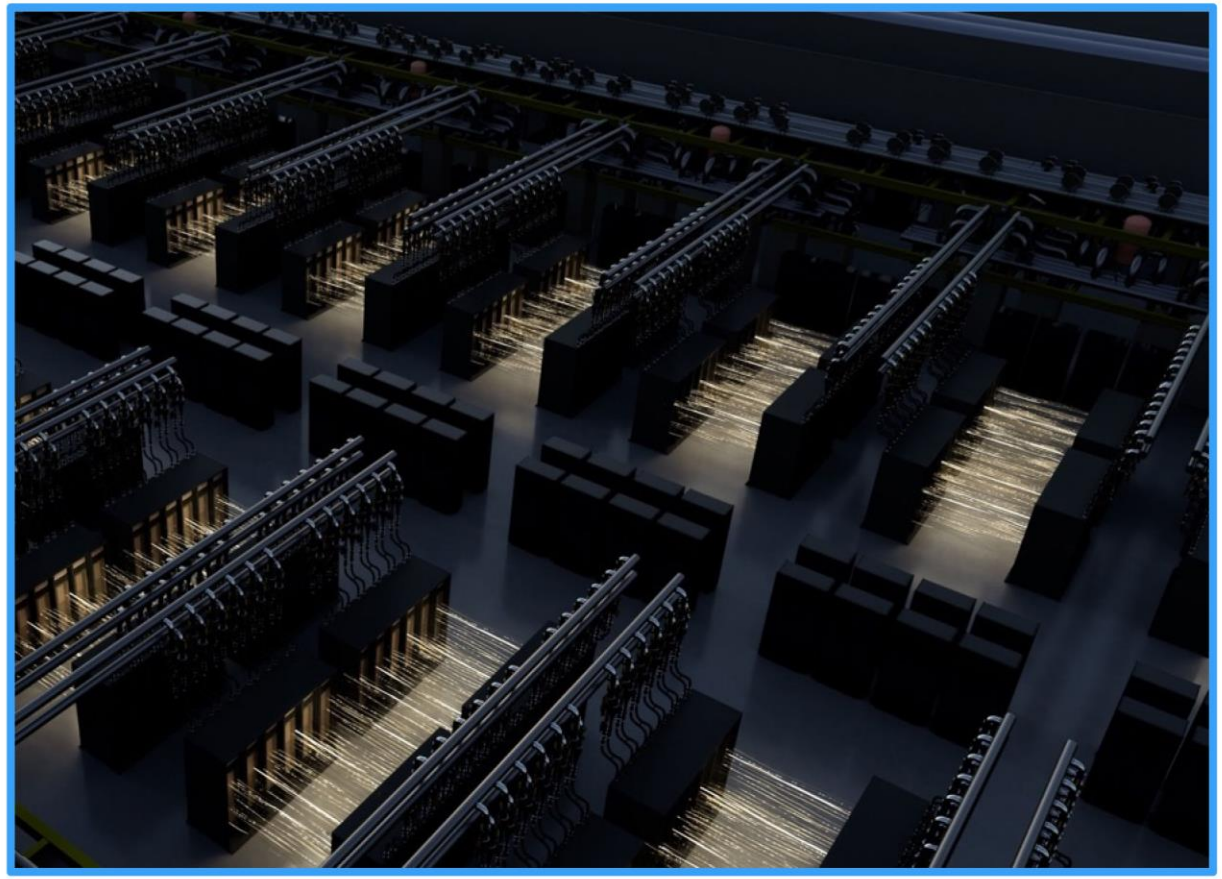
Two Trends Reshaping AI Data Center

Chip



Rack





Source: NVIDIA, "NVIDIA DSX AI Factory Platform".

Together, density and scale change the engineering requirements.

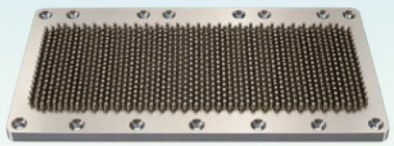
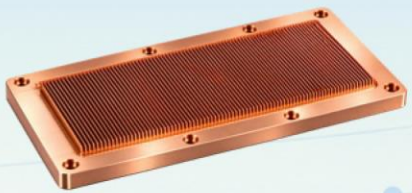
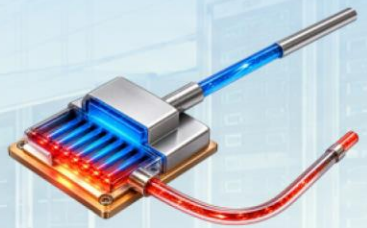
Higher Density is Changing Cooling Technology

Cooling architectures evolve as chip heat flux and system requirements increase

Higher Chip Heat Flux

Rising Thermal Requirements

Specialized Cooling Architecture

Single-Phase Cold Plate	Pin-Fin Cold Plate	Microchannel Cold Plate	Impingement Cold Plate	Two-Phase Cold Plate
				

We have to follow those changes back through the pumps, CDU and the facility cooling system.

Cooling Infrastructure Must Keep Pace

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Cooling architectures evolve as chip heat flux and system requirements increase

Heat Rejection Capacity
Enabling higher density AI infrastructure

300kW → 1500kW → 2500kW

Hydraulic Capability
Supporting higher flow Resistance

30psi → 60psi → 90psi

1.0LPM/kW → 1.5LPM/kW → 2.0LPM/kW



Capacity Change
Larger heat exchanger and pumps

Hydraulic Change
More powerful pumps, higher-rated pipes, valves and controls

Operating Flexibility
Coolant compatibility, temperature range and deployment conditions

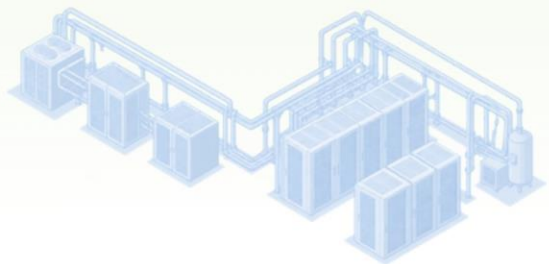
System Engineering Requires Deep Component Expertise

A component may meet its data sheet specification, how about the behavior in particular application.

Before

Select Mature Product

Integrate Known Interfaces



Now

Define Requirements → Co-develop Interfaces → Assess failure mode → Design Validation → Plan lifecycle support



In-depth analysis of design, materials, and operating behavior to fully understand component characteristics.



Rigorous reliability tests to ensure long-term durability under real-world operating conditions.



Evaluation of maintainability, serviceability, and lifecycle cost to ensure total ownership value.

The integrator's role has shifted from product selection to system validation.

At Scale, Small Variations becomes System Risk

Scale accelerates failure accumulation by repeating small manufacturing and installation variations across more units and operating scenarios.

Component



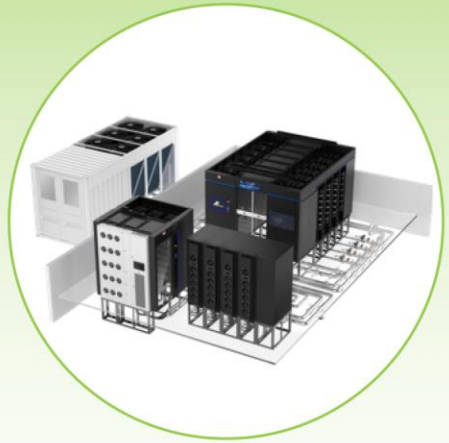
- Manufacturing Tolerance
- Seal and connector fit
- Cleanliness and material

Rack



- Repeated Interface
- Flow distribution difference
- Assembly and installation error

Facility



- Diverse Configurations
- Commissioning Difference
- Service and maintenance variation

Two things matters: the condition we start with, and what happens to that condition over time.

The Foundations of Long-Term Reliability

Long term reliability rests on manufacturing consistency before deployment and lifecycle control after deployment.

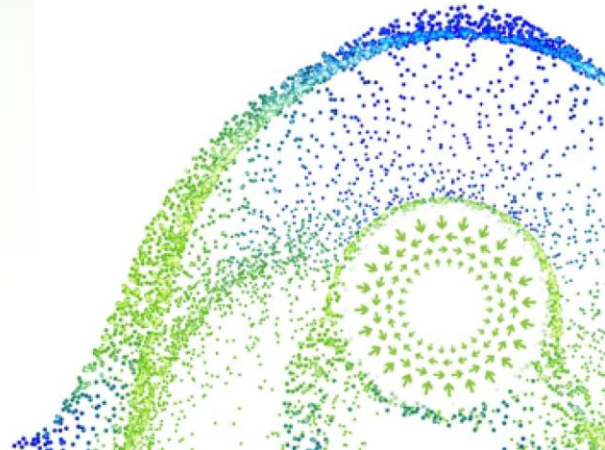
Long-term reliability at Scale

Manufacturing Consistency

Control variation before deployment

Lifecycle Control

Preserve condition throughout operation



Manufacturing Consistency

Manufacturing consistency begins with shared specifications and ends with repeatable, traceable results.

Shared standards

Common specifications
Unified acceptance criteria

Repeatable execution

Automated production
Controlled parameter process

Verified results

Recorded test result
Batch and unit traceability







The story doesn’t end at the factory gate.

Reliability must be managed across the lifecycle

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Feedback & Improvement
Capture operating experience
Analyze and improve
Feed into next generation



Production
Controlled process
Consistent quality
Built for reproducibility

Operation & Service
Stable and efficient operation
Preventive maintenance
Issue resolution

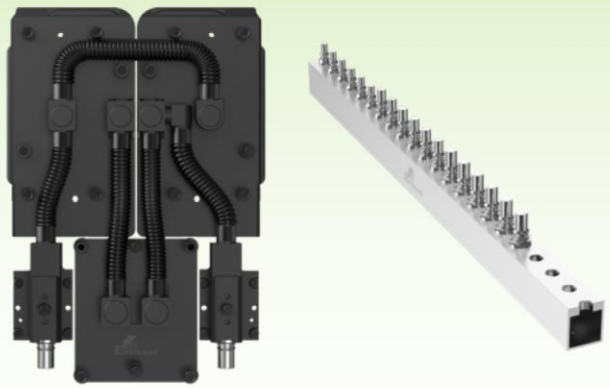
Installation & Commissioning
Proper installation
Functional verification
Ready for operation

Field experience returns to engineering and manufacturing

Lifecycle Cleanliness Control

Production

Prevent Contamination



- Controlled assembly environment
- Defined cleaning and welding
- Outgoing cleanliness check

Commissioning

Remove Installation Residue



- Defined flushing boundaries
- Controlled flushing procedure
- Verified acceptance criteria

Operation

Maintain Fluid Condition

Flow Control
Manage velocity to limit erosion

Filtration Performance
Evaluate beta ratio

Side-stream Filtration
Apply fine filtration where required

Fluid Compatibility
Check coolant & wetted materials

Online Monitoring
Track changes in fluid condition

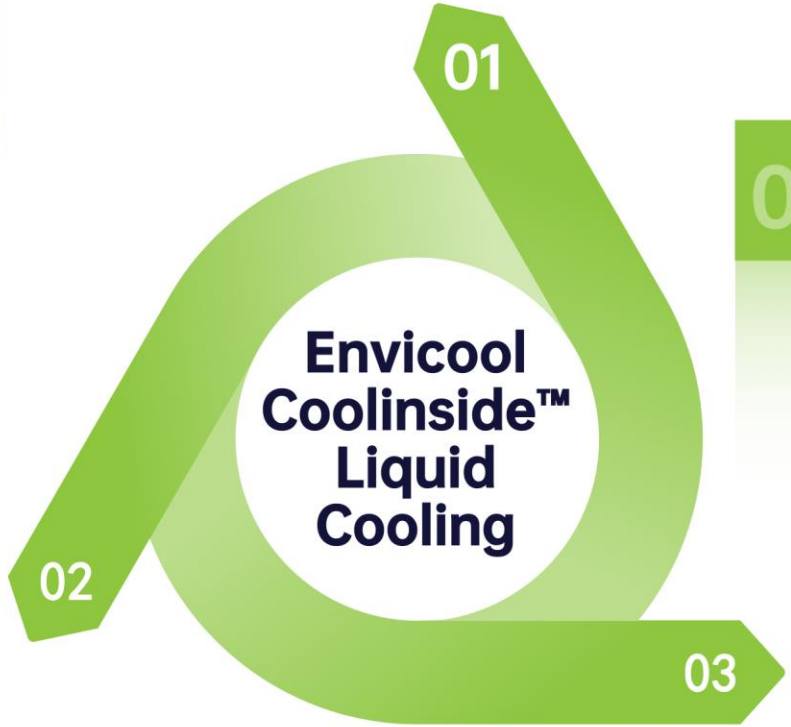
Envicool Full Chain Cooling Solution

01 Full Ecosystem Integration

From liquid server manufacturer to server-rack-pod-DC cooling /testing/maintenance

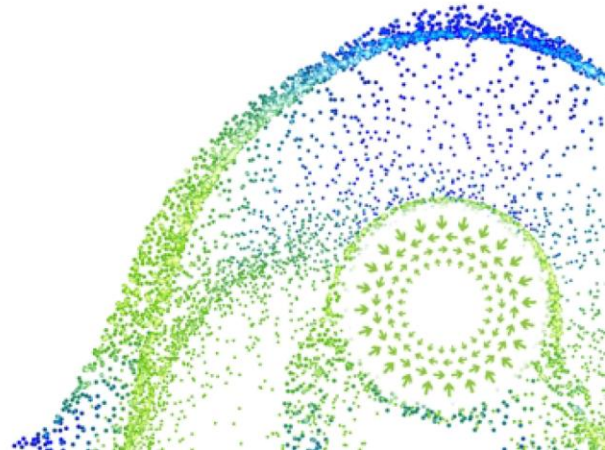
02 Full Application Scenario

Supporting kits and solutions to help deploy liquid cooling

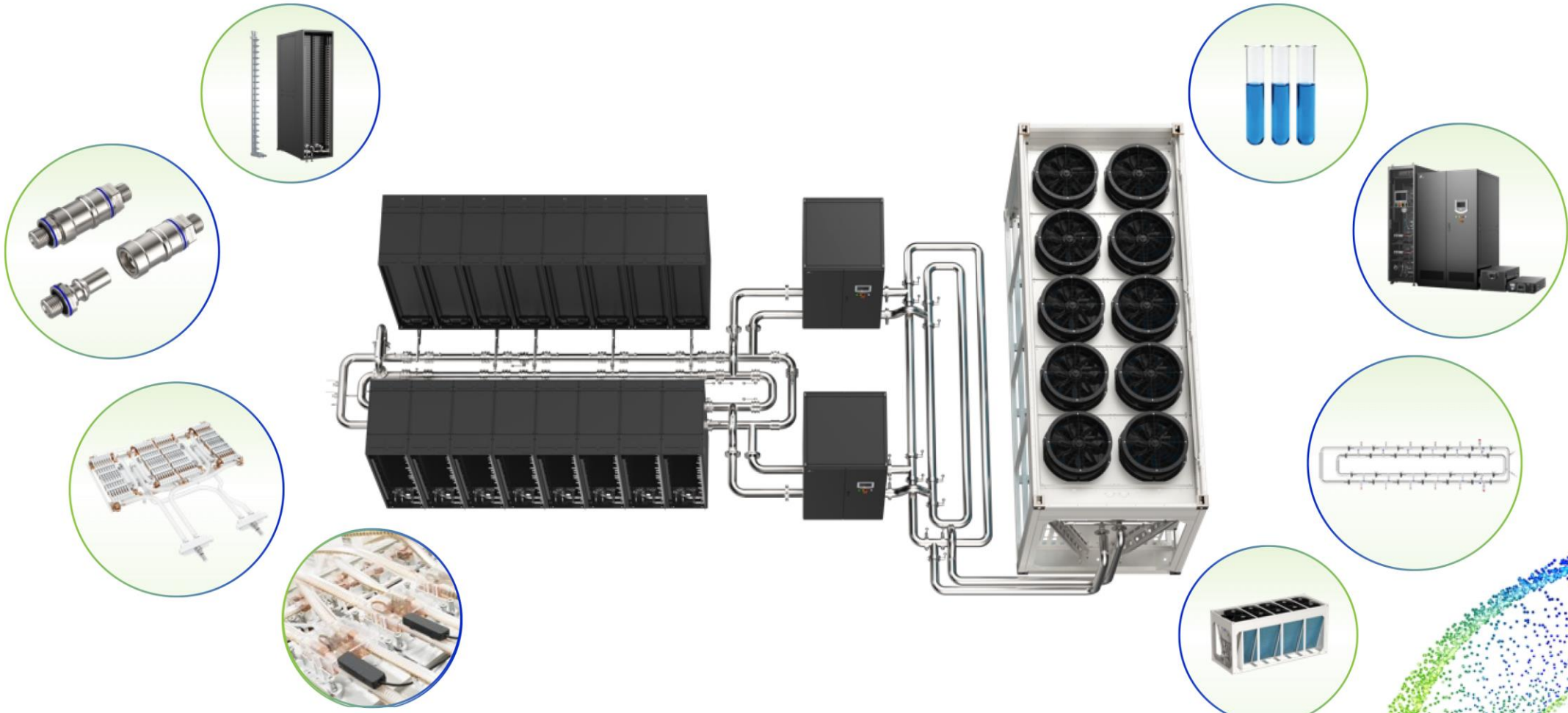


03 Full Lifecycle Management

Services to accelerate product development and field application



Full Eco-system Integration



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Full Application Coverage

Validation & Production

Lab Test



Production Line



DC Commissioning / O&M



Full Lifecycle Service

9 **Regional Hubs**
Fully owned strategic logistics & core regional offices.

22 **Service Networks**
Localized customer care, certified partner and storage facilities.

4M+ **Products Applied**
Proven experience in global delivery and end-to-end support.



Design >> Manufacturing >> Installation >> Maintenance

Welcome to Our Booth B19



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